SMART IMAGE ENHANCEMENT & ANALYSIS SYSTEM

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Objective

The objective of this project is to design and develop a smart image enhancement system using MATLAB. The system improves the quality of low-resolution and low-contrast images by applying various image processing techniques such as sampling, transformations, intensity adjustments, and histogram processing.

Methodology

The project is implemented in a structured pipeline:

Input Image → Analysis → Transformation → Enhancement → Output

Image Acquisition & Understanding

- The input image is loaded using MATLAB.
- If the image is RGB, it is converted to grayscale.
- Basic properties are analyzed:
    - Resolution (height × width)
    - Data type (uint8)

Observation:
The grayscale image simplifies processing and reduces complexity.

Sampling & Quantization

Sampling:

- Image is resized to:
    - 0.25x, 0.5x (down-sampling)
    - 1.5x, 2x (up-sampling)

Quantization:

- Bit depth reduced to:
    - 8-bit (original)
    - 4-bit
    - 2-bit

Observation:

- Down-sampling reduces resolution and detail
- Up-sampling increases size but may blur image
- Lower bit depth reduces image quality and introduces banding

Geometric Transformations

- Rotation applied at:
  - 30°, 45°, 60°, 90°, 120°, 150°, 180°
- Translation shifts the image position
- Shearing distorts the image shape

Observation:
Transformations change spatial structure. Rotation is reversible, but some transformations may cause information loss.

Intensity Transformations

Applied:

- **Negative Transformation** → highlights details
- **Log Transformation** → enhances dark regions
- **Gamma Correction:**
  - $\gamma = 0.5$ → brightens image
  - $\gamma = 1.5$ → darkens image

Observation:
Gamma correction ($\gamma < 1$) is best for improving brightness.

Histogram Processing

- Histogram of original image plotted
- Histogram Equalization applied

Observation:
Histogram equalization spreads intensity values, improving contrast and visibility.

Final Integrated System

Pipeline applied:

1. Gamma Correction ($\gamma = 0.5$)
2. Histogram Equalization

3. Gaussian Smoothing
4. Unsharp Masking

Final Output:
Enhanced image with improved contrast, brightness, and sharpness.

Code Snippet

```
img = imread('ZGa9z.jpg');
enhanced_image = process_image(img);
imshow(enhanced_image);
imwrite(enhanced_image, 'enhanced_output.png');
```

Input & Output

Input Image

Output Image

5. Observations & Analysis

- Histogram equalization significantly improves contrast
- Gamma correction helps in brightness adjustment
- Quantization reduces image quality
- Transformations affect spatial structure
- Combination of methods gives best results

Answers to Questions

Q1: Why does histogram equalization improve contrast?

It redistributes pixel intensity values across the full range, making dark areas lighter and bright areas clearer.

Q2: How does gamma affect brightness?

- $\gamma < 1 \rightarrow$ Brightens image
- $\gamma > 1 \rightarrow$ Darkens image

Q3: What is the effect of quantization on image quality?

Reducing bit depth decreases image quality and introduces visible intensity steps.

Q4: Which transformation is reversible and why?

Rotation (especially multiples of 90°) is reversible because no information is lost.

Q5: How do transformations affect spatial structure?

They change the position, orientation, and shape of objects in the image.